The early, unrealistic expectations resulted in a backlash; when everything did not immediately fall into place, many researchers were crestfallen. The string conferences of the late 1980s reflected the low-level disillusionment—physicists presented interesting results, but the atmosphere lacked inspiration. Some even suggested that the community stop holding an annual strings conference. But things picked up in the early 1990s. Through various breakthroughs, some of which we have discussed in previous chapters, string theory was rebuilding its momentum and researchers were regaining their excitement and optimism. But very little presaged what happened at the strings conference in March 1995 at the University of Southern California.

When his appointed hour to speak had arrived, Edward Witten strode to the podium and delivered a lecture that ignited the second superstring revolution. Inspired by earlier works of Duff, Hull, Townsend, and building on insights of Schwarz, the Indian physicist Ashok Sen, and others, Witten announced a strategy for transcending the perturbative understanding of string theory. A central part of the plan involves the concept of duality.

Physicists use the term duality to describe theoretical models that appear to be different but nevertheless can be shown to describe exactly the same physics. There are “trivial” examples of dualities in which ostensibly different theories are actually identical and appear to be different only because of the way in which they happen to be presented. To someone who knows only English, general relativity might not immediately be recognized as Einstein’s theory if presented in Chinese. A physicist fluent in both languages, though, can easily perform a translation from one to the other, establishing their equivalence. We call this example “trivial” because nothing is gained, from the point of view of physics, by such a translation. If someone who is fluent in English and Chinese were studying a difficult problem in general relativity, it would be equally challenging regardless of the language used to expressed it. A switch from English to Chinese, or vice versa, brings no new physical insight.

Nontrivial examples of duality are those in which distinct descriptions of the same physical situation do yield different and complementary physical insights and mathematical methods of